Durham University

Examination Paper

Examination Session:	Year:	Exam Code:
May/June	2022	ENGI3411-WE01

Title: (BEng/MEng)

Solid Mechanics 3

Time Allowed:	2 hours
Additional Material provided:	Yes, formula sheet included as part of exam paper
Reporting problems during the exam:	Queries about the instructions for this exam (rubric) should be directed to: eng.boe@durham.ac.uk. If you experience any difficulties uploading your exam answers to Gradescope report these immediately to: eng.boe@durham.ac.uk.

Instructions to Candidates:	Answer ALL questions. All questions carry equal marks. Please answer each question on a new page. All workings must be shown. This is an online open-book exam, submission is time-limited to 4 hours within a 24 hrs window. • *The examination attempt must be all your own work. You must not interact with another person (either one-to-one, by telephone, on the internet, in chat rooms, on social media, via smart watches, etc.), or use cheat sites, etc. whilst completing the exam. You must not discuss any examination papers with friends, relatives, current or past students of this or any other University before the end of the relevant 24-hour window.* • *Candidates are expected to answer all questions in their own words. Should close paraphrasing, direct quotation be necessary candidates must acknowledge the source of any material used in their answers (this includes text, images, diagrams, charts, tables and/or graphs). Full referencing is not expected, but a brief indication of the source of the material (e.g. name of author) must be given.*

Revision:	

Q1

You are called in to advise a small company that manufactures specialist gym equipment. Several customers have reported cracks developing in an alloy plate, and in some cases these have resulted in a fracture failure. The cracks have formed symmetrically from opposing edges in the manner shown in Figure Q1, and in the notation of the Figure the plate's dimensions are $h = 500$mm, $b = 120$mm. Unfortunately, their employee who did the initial fatigue analysis has left the company and no expertise remains in this area. The company provides you with the initial fatigue design calculations, which are not well documented but show the part to be subjected to a maximum stress of $\sigma = 36$ MPa (as shown in Figure Q1) with a stress ratio of -0.5. They cannot find reliable material data, but provide you with sample plates that suffered a fracture failure.

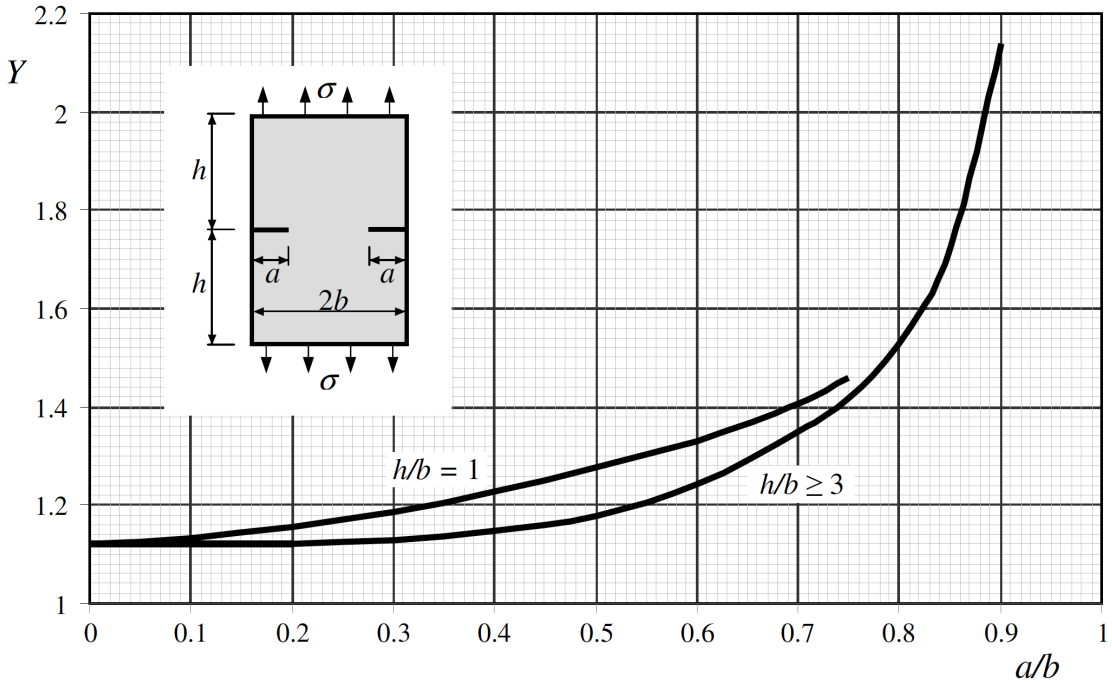


Figure Q1: Graph of Y factor for opposing edge cracks in strip

(a) You examine the failed specimens and find the cracks had reached a length of 85mm at fracture, and you see no evidence of significant yielding of material. Neglecting plastic zone correction, make a determination of the material's fracture toughness. **[10%]**

(b) Do a (quick) calculation to give a lower-bound value of the yield stress of the alloy. **[10%]**

(c) Several customers have reported cracks of length 70mm. The company urgently needs to predict the remaining number of load cycles so they can advise on safe usage while replacement parts are shipped and fitted. Fortunately they have found some more material data relating to crack growth rates, da/dN, with the range of stress intensity factors:

ΔK (MPa$\sqrt{\text{m}}$)	da/dN (m/cycle)
30	3.0×10^{-7}
50	1.2×10^{-6}

From Figure Q1 you find that Y is not constant over the crack growth. Using a straight line fit to the appropriate curve in the Figure, and stating key assumptions, estimate the remaining cycles to failure once the cracks have reached 70mm in length. **[50%]**

continued

(d) The Engineering Director of the company emails you as follows:

"We have found some more paperwork relating to the initial fatigue design. It was done to something called the Soderberg criterion. The engineer used a fatigue limit of 35MPa. Could you explain how this works? Please use simple terms - pretend I am an interested 14-year-old! I think you might need a yield stress value, too, but I cannot find that and it might take some time to find it. I will look some more during the Christmas break and let you know. Could you also speculate on the underlying reason these parts are suffering these fatigue failures?" .

Using your own words in your description, how might you respond? **[30%]**

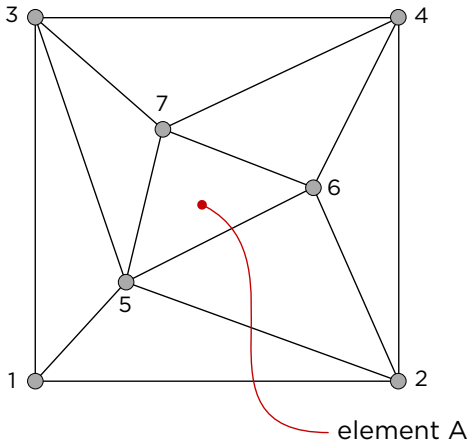
Q2

When developing finite element methods we make use of various techniques to verify and validate the numerical implementations and their ability to represent the physical problem we are attempting to analyse. This question centres on such techniques.

(a) *Patch tests* are used to confirm that elements satisfy completeness and compatibility requirements. In your own words, explain the implications of the compatibility requirement on the strain variation over a problem. **[10%]**

Consider the finite element problem shown in Figure Q2, which has been used to perform a patch test. The shape functions for the three-noded triangle are

$$N_1 = 1 - \xi - \eta, \qquad N_2 = \eta \qquad \text{and} \qquad N_3 = \xi.$$



node, i	x_i	y_i
1	0.000	0.000
2	1.000	0.000
3	0.000	1.000
4	1.000	1.000
5	0.290	0.318
6	0.765	0.533
7	0.409	0.810

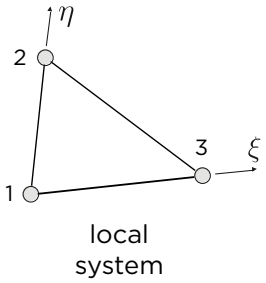


Figure Q2: finite element mesh, nodal coordinates & local system where $\xi, \eta \in [0, 1]$.

(b) If the displacement vector for global nodes 5, 6 and 7 is

$$\{d\} = \{u_5 \quad v_5 \quad u_6 \quad v_6 \quad u_7 \quad v_7\}^T = \{0.00 \quad 0.12 \quad 0.05 \quad 0.08 \quad 0.07 \quad 0.02\}^T$$

determine the strain state in element A. **[50%]**

One way to confirm that a finite element approach has been implemented correctly is to conduct a convergence study using a problem where we know the exact solution. However, in some cases exact solutions do not exist and therefore we must use other methods to confirm that a code is correct. One such technique is the *method of manufactured solutions* where we start from an assumed displacement field, calculate the required body forces and tractions to generate such a displacement field, impose these loads on the finite elements and check the resulting displacement solution converges towards the assumed solution with mesh refinement.

(c) Assuming that the displacement field for a 2D plane strain problem is

$$\{u\} = \begin{Bmatrix} \sin(2\pi x) \\ \sin(2\pi y) \end{Bmatrix}$$

use the strong form statement of equilibrium for 2D stress analysis,

$$\frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \sigma_{xy}}{\partial y} + f_x^b = 0 \qquad \text{and} \qquad \frac{\partial \sigma_{yx}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} + f_y^b = 0,$$

to determine the **body forces** per unit volume that need to be imposed to cause this displacement. Show all of your steps and clearly state any assumptions that you make. **[40%]**

continued

Q3

Consider the physical problem shown in Figure Q3(a) where a pin spanner is being used to tighten a lockring.

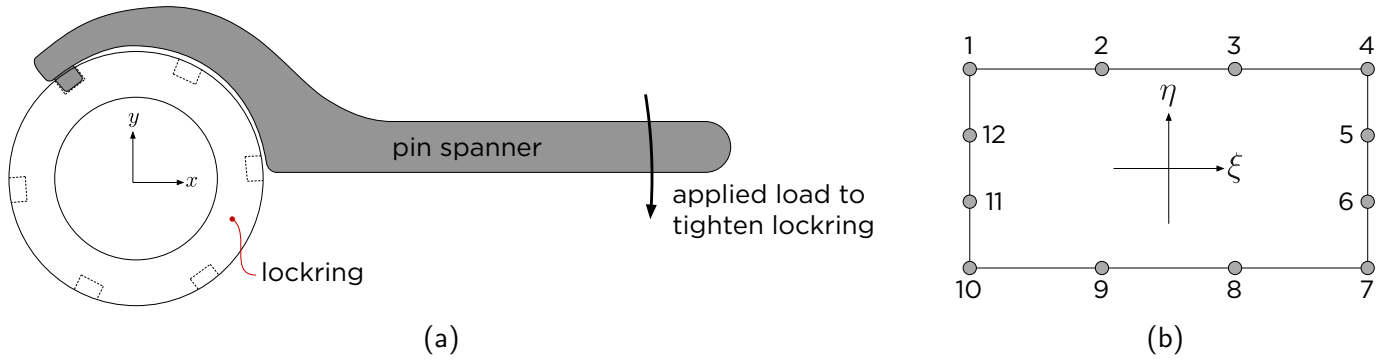


Figure Q3: (a) pin spanner & lockring and (b) local node numbering where $(\xi, \eta) \in [-1, 1]$

When answering the following questions, assume that you would only be analysing the pin spanner and trying to understand the stress state through the object.

(a) If analysing this as a 2D problem, what assumption would you assume in the third direction? Justify your answer with physical arguments. **[10%]**

(b) Explain the boundary conditions that you would impose on the spanner in a finite element analysis. Produce a hand drawn sketch to clearly identify the boundary conditions on different parts of the spanner. **[20%]**

(c) If you only had a finite number of elements to analyse the problem, explain how you would distribute the elements within the analysis. Use a hand drawn sketch to support your answer. **[15%]**

The weak form statement of equilibrium for finite element analysis can be expressed as

$$\int_V [B]^T [D^e][B] dV \ \{d\} = \int_S [N]^T \{t\} dS + \int_V [N]^T \{f^b\} dV.$$

It is standard to evaluate the above integrals using Gauss-Legendre quadrature.

(d) If you were analysing a problem using 12-noded bi-cubic quadrilateral elements with regular shapes, what order of quadrature would you use to evaluate the stiffness integral in the equilibrium statement? Would your answer change if the elements were not regular? Justify your answers. **[15%]**

Consider the 12-noded bi-cubic quadrilateral element shown in Figure Q3(b). The element has a physical length of 25mm and 12.5mm in the x and y directions and the global and local coordinates are aligned. The element is loaded over its top surface via a constant surface traction of 80kPa acting vertically downwards. The shape functions for local nodes 1 and 2 are

$$N_1 = -\frac{1}{32}(\xi - 1)(\eta + 1)(9\xi^2 + 9\eta^2 - 10) \qquad \text{and} \qquad N_2 = \frac{9}{32}(\xi - 1)(\eta + 1)(\xi + 1)(3\xi - 1)$$

(e) Determine the equivalent nodal load at nodes 1 and 2 due to the imposed traction. **[40%]**

ENGI3411-WE01 Solid Mechanics 3
Course: Fracture and Fatigue
Formula sheet

Stress and displacement distribution locally around the tip of a mode I crack oriented in the x-direction

$$\sigma_{xx} = \frac{K_I}{\sqrt{2\pi r}} \cos\frac{\theta}{2}\left[1 - \sin\frac{\theta}{2}\sin\frac{3\theta}{2}\right] + ...$$

$$\sigma_{yy} = \frac{K_I}{\sqrt{2\pi r}} \cos\frac{\theta}{2}\left[1 + \sin\frac{\theta}{2}\sin\frac{3\theta}{2}\right] + ...$$

$$\sigma_{xy} = \frac{K_I}{\sqrt{2\pi r}} \sin\frac{\theta}{2}\cos\frac{\theta}{2}\cos\frac{3\theta}{2} + ...$$

$$u_x = \frac{K_\mathrm{I}}{\mu}\sqrt{\frac{r}{2\pi}}\cos\frac{\theta}{2}\left[\frac{1}{2}(\kappa - 1) + \sin^2\frac{\theta}{2}\right] + \cdots$$

$$u_y = \frac{K_\mathrm{I}}{\mu}\sqrt{\frac{r}{2\pi}}\sin\frac{\theta}{2}\left[\frac{1}{2}(\kappa + 1) - \cos^2\frac{\theta}{2}\right] + \cdots$$

$\kappa = 3 - 4\nu$ (plane strain), or $(3-\nu)/(1+\nu)$ (plane stress)

Mode I stress intensity factor

$$K_I = Y\sigma\sqrt{\pi a}$$

Internal crack of length $2a$ in infinite plate: $Y = 1$
Edge crack of length a in relatively large plate: $Y = 1.12$

Plastic zone correction

$$\text{Equivalent crack length} = a + \frac{K_I^2}{2\pi\sigma_Y^2}$$

Failure diagrams for mean stress correction

Modified Goodman:

$$S_a = S_D\left[1 - \frac{S_{mean}}{S_u}\right]$$

Gerber parabola:

$$S_a = S_D\left[1 - \left(\frac{S_{mean}}{S_u}\right)^2\right]$$

Soderberg:

$$S_a = S_D\left[1 - \frac{S_{mean}}{S_{yield}}\right]$$

Strain-life method

Strain life equation:

$$\frac{\Delta\varepsilon}{2} = \frac{\sigma_f'}{E}(2N_f)^b + \varepsilon_f'(2N_f)^c$$

Mean stress correction (Smith Watson Topper):

$$\left(\sigma_{mean} + \frac{\Delta\sigma}{2}\right)\frac{\Delta\varepsilon}{2} = \frac{\sigma_f'^2}{E}(2N_f)^{2b} + \sigma_f'\varepsilon_f'(2N_f)^{b+c}$$

continued

Macro crack growth laws based on fracture mechanics approach

Paris Law:

$$\frac{da}{dN} = C(\Delta K)^m$$

Broek and Schijve:

$$\frac{da}{dN} = CK_{max}^2 \Delta K$$

Walker:

$$\frac{da}{dN} = CK_{max}^m (\Delta K)^n$$

Walker:

$$\frac{da}{dN} = C\overline{\Delta K}^n \qquad \text{where } \overline{\Delta K} = S_{max}(1-R)^m\sqrt{\pi a}$$

Forman:

$$\frac{da}{dN} = \frac{C(\Delta K)^m}{(1-R)(K_{Ic} - K_{max})}$$

continued

ENGI3411-WE01 Solid Mechanics 3
Course: Finite-element method
Formula sheet

Three-dimensional stress-strain relationship

$$\begin{Bmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \sigma_{zz} \\ \sigma_{xy} \\ \sigma_{yz} \\ \sigma_{zx} \end{Bmatrix} = \frac{E}{(1+\nu)(1-2\nu)} \begin{bmatrix} (1-\nu) & \nu & \nu & 0 & 0 & 0 \\ \nu & (1-\nu) & \nu & 0 & 0 & 0 \\ \nu & \nu & (1-\nu) & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{(1-2\nu)}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{(1-2\nu)}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{(1-2\nu)}{2} \end{bmatrix} \begin{Bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \varepsilon_{zz} \\ \gamma_{xy} \\ \gamma_{yz} \\ \gamma_{zx} \end{Bmatrix}$$

Two-dimensional stress-strain relationships

$$\begin{Bmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \sigma_{xy} \end{Bmatrix} = \frac{E}{(1+\nu)(1-2\nu)} \begin{bmatrix} (1-\nu) & \nu & 0 \\ \nu & (1-\nu) & 0 \\ 0 & 0 & \frac{(1-2\nu)}{2} \end{bmatrix} \begin{Bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \gamma_{xy} \end{Bmatrix} \qquad \text{(plane strain)}$$

$$\begin{Bmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \sigma_{xy} \end{Bmatrix} = \frac{E}{1-\nu^2} \begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{(1-\nu)}{2} \end{bmatrix} \begin{Bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \gamma_{xy} \end{Bmatrix} \qquad \text{(plane stress)}$$

Three-dimensional strain definition

$$\begin{Bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \varepsilon_{zz} \\ \gamma_{xy} \\ \gamma_{yz} \\ \gamma_{zx} \end{Bmatrix} = [L]\{u\} = \begin{bmatrix} \frac{\partial}{\partial x} & 0 & 0 \\ 0 & \frac{\partial}{\partial y} & 0 \\ 0 & 0 & \frac{\partial}{\partial z} \\ \frac{\partial}{\partial y} & \frac{\partial}{\partial x} & 0 \\ 0 & \frac{\partial}{\partial z} & \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} & 0 & \frac{\partial}{\partial x} \end{bmatrix} \begin{Bmatrix} u \\ v \\ w \end{Bmatrix}$$

Interpolation using shape functions

$$u = \sum_{i=1}^{n_{en}} N_i u_i = \{N\}^T \{u\}$$

Three-dimensional strain-displacement definition

$$\begin{Bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \varepsilon_{zz} \\ \gamma_{xy} \\ \gamma_{yz} \\ \gamma_{zx} \end{Bmatrix} = \underbrace{\begin{bmatrix} \frac{\partial N_1}{\partial x} & 0 & 0 & \dots & \frac{\partial N_{n_{en}}}{\partial x} & 0 & 0 \\ 0 & \frac{\partial N_1}{\partial y} & 0 & \dots & 0 & \frac{\partial N_{n_{en}}}{\partial y} & 0 \\ 0 & 0 & \frac{\partial N_1}{\partial z} & \dots & 0 & 0 & \frac{\partial N_{n_{en}}}{\partial z} \\ \frac{\partial N_1}{\partial y} & \frac{\partial N_1}{\partial x} & 0 & \dots & \frac{\partial N_{n_{en}}}{\partial y} & \frac{\partial N_{n_{en}}}{\partial x} & 0 \\ 0 & \frac{\partial N_1}{\partial z} & \frac{\partial N_1}{\partial y} & \dots & 0 & \frac{\partial N_{n_{en}}}{\partial z} & \frac{\partial N_{n_{en}}}{\partial y} \\ \frac{\partial N_1}{\partial z} & 0 & \frac{\partial N_1}{\partial x} & \dots & \frac{\partial N_{n_{en}}}{\partial z} & 0 & \frac{\partial N_{n_{en}}}{\partial x} \end{bmatrix}}_{[B]=[L][N]} \begin{Bmatrix} u_1 \\ v_1 \\ w_1 \\ \vdots \\ u_{n_{en}} \\ v_{n_{en}} \\ w_{n_{en}} \end{Bmatrix}$$

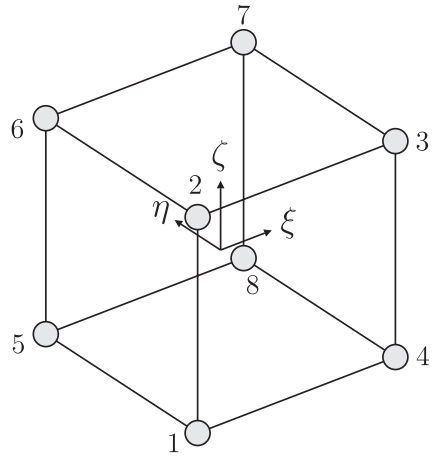
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Plane-stress/strain strain-displacement definition

$$\left\{\begin{array}{c}\varepsilon_{xx}\\ \varepsilon_{yy}\\ \gamma_{xy}\end{array}\right\} = \left[\begin{array}{cccc}\frac{\partial N_1}{\partial x} & 0 & \cdots & \frac{\partial N_{n_{en}}}{\partial x} & 0\\ 0 & \frac{\partial N_1}{\partial y} & \cdots & 0 & \frac{\partial N_{n_{en}}}{\partial y}\\ \frac{\partial N_1}{\partial y} & \frac{\partial N_1}{\partial x} & \cdots & \frac{\partial N_{n_{en}}}{\partial y} & \frac{\partial N_{n_{en}}}{\partial x}\end{array}\right] \left\{\begin{array}{c}u_1\\ v_1\\ \vdots\\ u_{n_{en}}\\ v_{n_{en}}\end{array}\right\}.$$

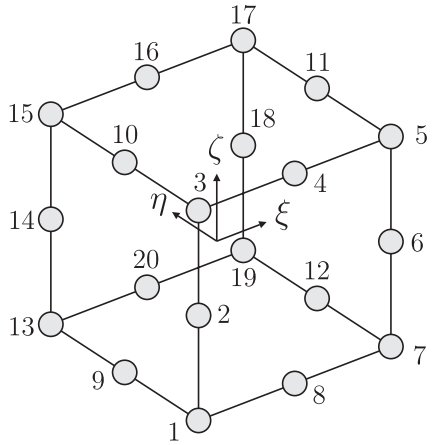
Shape functions for hexahedral elements

Tri-linear eight-noded hexahedral



$$\begin{aligned}
N_1 &= (1-\xi)(1-\eta)(1-\zeta)/8\\
N_2 &= (1-\xi)(1-\eta)(1+\zeta)/8\\
N_3 &= (1+\xi)(1-\eta)(1+\zeta)/8\\
N_4 &= (1+\xi)(1-\eta)(1-\zeta)/8\\
N_5 &= (1-\xi)(1+\eta)(1-\zeta)/8\\
N_6 &= (1-\xi)(1+\eta)(1+\zeta)/8\\
N_7 &= (1+\xi)(1+\eta)(1+\zeta)/8\\
N_8 &= (1+\xi)(1+\eta)(1-\zeta)/8
\end{aligned}$$

Tri-quadratic twenty-noded hexahedral



$$\begin{aligned}
N_1 &= (1-\xi)(1-\eta)(1-\zeta)(-\xi-\eta-\zeta-2)/8\\
N_2 &= (1-\xi)(1-\eta)(1-\zeta^2)/4\\
N_3 &= (1-\xi)(1-\eta)(1+\zeta)(-\xi-\eta+\zeta-2)/8\\
N_4 &= (1-\xi^2)(1-\eta)(1+\zeta)/4\\
N_5 &= (1+\xi)(1-\eta)(1+\zeta)(\xi-\eta+\zeta-2)/8\\
N_6 &= (1+\xi)(1-\eta)(1-\zeta^2)/4\\
N_7 &= (1+\xi)(1-\eta)(1-\zeta)(\xi-\eta-\zeta-2)/8\\
N_8 &= (1-\xi^2)(1-\eta)(1-\zeta)/4\\
N_9 &= (1-\xi)(1-\eta^2)(1-\zeta)/4\\
N_{10} &= (1-\xi)(1-\eta^2)(1+\zeta)/4\\
N_{11} &= (1+\xi)(1-\eta^2)(1+\zeta)/4\\
N_{12} &= (1+\xi)(1-\eta^2)(1-\zeta)/4\\
N_{13} &= (1-\xi)(1+\eta)(1-\zeta)(-\xi+\eta-\zeta-2)/8\\
N_{14} &= (1-\xi)(1+\eta)(1-\zeta^2)/4\\
N_{15} &= (1-\xi)(1+\eta)(1+\zeta)(-\xi+\eta+\zeta-2)/8\\
N_{16} &= (1-\xi^2)(1+\eta)(1+\zeta)/4\\
N_{17} &= (1+\xi)(1+\eta)(1+\zeta)(\xi+\eta+\zeta-2)/8\\
N_{18} &= (1+\xi)(1+\eta)(1-\zeta^2)/4\\
N_{19} &= (1+\xi)(1+\eta)(1-\zeta)(\xi+\eta-\zeta-2)/8\\
N_{20} &= (1-\xi^2)(1+\eta)(1-\zeta)/4
\end{aligned}$$

continued

Axi-symmetric strain-displacement definition

$$\left\{\begin{array}{c}\varepsilon_{rr}\\ \varepsilon_{zz}\\ \varepsilon_{\theta\theta}\\ \gamma_{rz}\end{array}\right\}=\left[\begin{array}{ccccc}\frac{\partial N_1}{\partial r} & 0 & \cdots & \frac{\partial N_{nen}}{\partial r} & 0\\ 0 & \frac{\partial N_1}{\partial z} & \cdots & 0 & \frac{\partial N_{nen}}{\partial z}\\ \frac{N_1}{r_1} & 0 & \cdots & \frac{N_{nen}}{r_{nen}} & 0\\ \frac{\partial N_1}{\partial z} & \frac{\partial N_1}{\partial r} & \cdots & \frac{\partial N_{nen}}{\partial z} & \frac{\partial N_{nen}}{\partial r}\end{array}\right]\left\{\begin{array}{c}u_1\\ w_1\\ \vdots\\ u_{nen}\\ w_{nen}\end{array}\right\}.$$

Global shape function derivatives

$$\left\{\begin{array}{c}\frac{\partial N_i}{\partial x}\\ \frac{\partial N_i}{\partial y}\\ \frac{\partial N_i}{\partial z}\end{array}\right\}=\left[\begin{array}{ccc}\frac{\partial \xi}{\partial x} & \frac{\partial \eta}{\partial x} & \frac{\partial \zeta}{\partial x}\\ \frac{\partial \xi}{\partial y} & \frac{\partial \eta}{\partial y} & \frac{\partial \zeta}{\partial y}\\ \frac{\partial \xi}{\partial z} & \frac{\partial \eta}{\partial z} & \frac{\partial \zeta}{\partial z}\end{array}\right]\left\{\begin{array}{c}\frac{\partial N_i}{\partial \xi}\\ \frac{\partial N_i}{\partial \eta}\\ \frac{\partial N_i}{\partial \zeta}\end{array}\right\}.$$

The Jacobian matrix

$$[J]=\left[\begin{array}{ccc}\frac{\partial x}{\partial \xi} & \frac{\partial y}{\partial \xi} & \frac{\partial z}{\partial \xi}\\ \frac{\partial x}{\partial \eta} & \frac{\partial y}{\partial \eta} & \frac{\partial z}{\partial \eta}\\ \frac{\partial x}{\partial \zeta} & \frac{\partial y}{\partial \zeta} & \frac{\partial z}{\partial \zeta}\end{array}\right]=\left[\begin{array}{cccc}\frac{\partial N_1}{\partial \xi} & \frac{\partial N_2}{\partial \xi} & \cdots & \frac{\partial N_{nen}}{\partial \xi}\\ \frac{\partial N_1}{\partial \eta} & \frac{\partial N_2}{\partial \eta} & \cdots & \frac{\partial N_{nen}}{\partial \eta}\\ \frac{\partial N_1}{\partial \zeta} & \frac{\partial N_2}{\partial \zeta} & \cdots & \frac{\partial N_{nen}}{\partial \zeta}\end{array}\right]\left[\begin{array}{ccc}x_1 & y_1 & z_1\\ x_2 & y_2 & z_2\\ \vdots & \vdots & \vdots\\ x_{n_{en}} & y_{n_{en}} & z_{n_{en}}\end{array}\right]$$

Weak form of equilibrium

$$\int_V [B]^T[D^{\mathsf{e}}][B]dV\{d\}=\int_S [N]^T\{t\}dS+\int_V [N]^T\{f^b\}dV.$$

Element stiffness and force vector

$$[k^e]=\int_V [B]^T[D^{\mathsf{e}}][B]dV \qquad \text{and} \qquad \{f^e\}=\int_S [N]^T\{t\}dS+\int_V [N]^T\{f^b\}dV.$$

Gauss-Legendre quadrature

$$\int_{-1}^{1} f(x)dx \approx \sum_{i=1}^{n_{Gp}} f(x_i)w_i$$

n_{Gp}	Abscissae, x_i	Weights, w_i
1	0.000000000000000	2.000000000000000
2	±0.577350269189626	1.000000000000000
3	0.000000000000000	0.888888888888889
	±0.774596669241483	0.555555555555556
4	±0.339981043584856	0.652145154862546
	±0.861136311594053	0.347854845137454

Numerical approximation of the element stiffness matrix

$$[k^e] \approx \sum_{i=1}^{n_{Gp}} [B]_i^T[D^{\mathsf{e}}][B]_i w_i \det([J]_i)$$

END